# The Atomic Nature of Matter - Class Diagram

package Core

**Main**

+main(String[])

calls

**Manager**

+getAvogadro(int, double, double, String)

-trackPictures(...)

-getSortedPictures(String)

-findBeads(String, int, double)

«uses» via IFactory / interfaces

**Legend**

Core | Interfaces | Models | Utilities

implements | uses / depends on | creates

+ public - private Picture also implements java.awt.event.ActionListener

package Interfaces

«interface» **IFactory**

+createBlob(): IBlob

+createPicture(String): IPicture

+createCleaner(IPicture, double)

+createBeadFinder(boolean[][])

+createBeadTracker(f1, f2)

+createAvogadro(): IAvogadro

«interface» **IPicture**

+getLuminance(int, int): double

+width(): int

+height(): int

«interface» **IDataCleaner**

+getCleanedData(): boolean[][]

«interface» **IBeadFinder**

+getBeads(int): ArrayList<IBlob>

«interface» **IBlob**

+addPixel(int, int)

+mass(): int

+distance(IBlob): double

+getCenterX(): double

+getCenterY(): double

+toString(): String

«interface» **IBeadTracker**

+trackBeads(double, PrintWriter)

«interface» **IAvogadro**

+calculateBoltzmann(String)

+calculateAvogadro(String)

uses

uses

uses

**Picture**

(Princeton stdlib image class)

+getLuminance(int, int): double

+width(): int

+height(): int

**DataCleaner**

-picture: IPicture

-cleanedData: boolean[][]

-tau: double

+getCleanedData(): boolean[][]

**BeadFinder**

-cleanedData: boolean[][]

-visited: boolean[][]

-blobs: ArrayList<IBlob>

+getBeads(int): ArrayList<IBlob>

-DFS_Search(int, int)

**Blob**

-X: double

-Y: double

-mass: int

+addPixel(int, int)

+mass(): int

+distance(IBlob): double

+getCenterX(): double

+getCenterY(): double

+toString(): String

**BeadTracker**

-beadsF1: ArrayList<IBlob>

-beadsF2: ArrayList<IBlob>

+trackBeads(double, PrintWriter)

**Avogadro**

+calculateBoltzmann(String)

+calculateAvogadro(String)

-calculateVariance(String)

-calculateDiffusionConstant(...)

uses

**Config**

+PIXEL_SIZE = 0.175e-6

+DELTA_T = 0.5

+TEMPERATURE = 297

+VISCOSITY = 9.135e-4

+BEAD_RADIUS = 0.5e-6

+R = 8.31446

**«creates» (only IFactory instantiates the concrete classes)**

BeadFinder «uses» IFactory.createBlob()